

3.5 Quiz: Vectors, Frames of Reference, and Energy — Solutions

1. Matching: physicists and their work.

- Newton (1687) **(c)** — laws of motion and gravitation
Michelson & Morley (1887) **(a)** — interferometer, null result for ether
Poincaré (1900) **(e)** — principle of relativity, simultaneity
Lorentz (1904) **(d)** — transformation equations
Einstein (1905) **(b)** — constancy of c , special relativity

2. Effects on measurements at high relative speed.

- (a) **Length.** The object is measured to be *shorter* along the direction of relative motion (length contraction). Its measured length is $L = L_0/\gamma$, where L_0 is the proper (rest-frame) length and $\gamma = 1/\sqrt{1 - v^2/c^2} \geq 1$. Lengths perpendicular to the motion are unchanged.
- (b) **Time.** Moving clocks are measured to run *slower* (time dilation). A time interval measured in the moving frame as Δt_0 is measured as $\Delta t = \gamma \Delta t_0$ by the stationary observer — so more time passes for the observer than for the moving clock.
- (c) **Inertia.** The force required to produce a given acceleration *increases* with speed. The object's effective inertia behaves as γm_0 , growing without bound as $v \rightarrow c$. Because $\gamma \rightarrow \infty$, no finite force can accelerate an object with nonzero rest mass up to the speed of light.

3. **Force components** ($\vec{F} = 100 \text{ N}$ at 53° above horizontal).

$$(a) \quad F_x = F \cos \theta = (100 \text{ N}) \cos 53^\circ \approx (100)(0.602) \approx \boxed{60 \text{ N}}$$

$$(b) \quad F_y = F \sin \theta = (100 \text{ N}) \sin 53^\circ \approx (100)(0.799) \approx \boxed{80 \text{ N}}$$

$$\text{Check: } \sqrt{60^2 + 80^2} = \sqrt{3600 + 6400} = \sqrt{10000} = 100 \text{ N. } \checkmark$$

4. **Magnitude and direction** ($v_x = 8 \text{ m/s}$, $v_y = 6 \text{ m/s}$).

$$(a) \quad |\vec{v}| = \sqrt{v_x^2 + v_y^2} = \sqrt{8^2 + 6^2} = \sqrt{64 + 36} = \sqrt{100} = \boxed{10 \text{ m/s}}$$

$$(b) \quad \theta = \arctan\left(\frac{v_y}{v_x}\right) = \arctan\left(\frac{6}{8}\right) = \arctan(0.75) \approx 36.87^\circ \approx \boxed{37^\circ} \text{ north of east.}$$

5. **Frame of reference — train.**

(a) Take east as positive. The passenger walks forward at $+2.0 \text{ m/s}$ relative to the train, which moves at $+30 \text{ m/s}$ relative to the ground.

$$v_{\text{passenger/ground}} = v_{\text{passenger/train}} + v_{\text{train/ground}} = 2.0 + 30 = \boxed{32 \text{ m/s east}}$$

(b) The car moves at -20 m/s (west) relative to the ground.

$$v_{\text{passenger/car}} = v_{\text{passenger/ground}} - v_{\text{car/ground}} = 32 - (-20) = \boxed{52 \text{ m/s east}}$$

6. Frame of reference — airplane in wind.

The plane's velocity relative to the air is $(60, 0)$ m/s (east). The air's velocity relative to the ground is $(0, 25)$ m/s (north). Adding,

$$\vec{v}_{\text{plane/ground}} = (60, 25) \text{ m/s.}$$

$$(a) \quad |\vec{v}| = \sqrt{60^2 + 25^2} = \sqrt{3600 + 625} = \sqrt{4225} = \boxed{65 \text{ m/s}}$$

$$(b) \quad \theta = \arctan\left(\frac{25}{60}\right) \approx 22.62^\circ \approx \boxed{23^\circ} \text{ north of east.}$$

7. Energy unit conversions ($E = 4.0 \times 10^5$ J).

(a) Kilocalories:

$$E = 4.0 \times 10^5 \text{ J} \times \frac{1 \text{ kcal}}{4184 \text{ J}} \approx \boxed{96 \text{ kcal}}$$

(About 96 “Calories” — close to the food-label value for a medium banana.)

(b) Kilowatt-hours:

$$E = 4.0 \times 10^5 \text{ J} \times \frac{1 \text{ kWh}}{3.6 \times 10^6 \text{ J}} \approx \boxed{0.11 \text{ kWh}}$$

(c) Electron-volts:

$$E = 4.0 \times 10^5 \text{ J} \times \frac{1 \text{ eV}}{1.6 \times 10^{-19} \text{ J}} = \boxed{2.5 \times 10^{24} \text{ eV}}$$